MEHMET BAHA DURSUN

Robotics Engineer | baha.dursun123@gmail.com | Odense, Denmark | D-type visa, Work permit 90 hours
Portfolio: renbago.github.io/MehmetBahaDursun | GitHub: github.com/Renbago | LinkedIn: linkedin.com/in/mehmet-baha-dursun

PROFESSIONAL SUMMARY

Robotics engineer whose localization and obstacle-detection systems run on 400+ deployed service robots, now moving into robot learning and manipulation. I built the core 2D/3D localization stack at Saha Robotik and now lead the company's robot-learning work (LeRobot and vision-language-action policies), working remotely from Denmark. MSc student in Advanced Robotics at the University of Southern Denmark, where I built a diffusion-policy pipeline on the physical da Vinci Research Kit as a part-time student research assistant. Experienced in ROS 2/ROS, imitation learning, and team leadership.

PROFESSIONAL EXPERIENCE

Robotics Engineer | Saha Robotik, Istanbul, Turkey | November 2023 - Present

- Designed and built the core localization and mapping architecture for 400+ autonomous waiter/cargo robots operating 24/7 in commercial environments.
- Managed complete project lifecycle for Turkey's first 24/7 autonomous airport guide robot at Istanbul Airport, covering 137,221 m² with 25 km minimum daily travel. (video, project page)
- Since relocating to Denmark, work remotely: still lead the 2D/3D localization work, now in a more supervisory role as the stack has matured, and lead the company's new bimanual mobile-manipulation project for restaurant table clearing (OpenArm hardware, teleoperation, data collection, vision-language-action / LeRobot policies), toward a planned Eurostars collaboration between SDU and Saha. (project page)
- Built the dynamic and static obstacle-detection algorithm alongside the localization stack.
- Led Saha's ROS training program for university students in Azerbaijan (February - April 2025): one month of remote teaching, then on-site practical sessions and evaluation of the students' final projects. (program post, project page)
- Technologies: ROS 2, C/C++, Python, SLAM, 2D/3D Localization, Obstacle Detection.

Student Research Assistant (Part-Time) | University of Southern Denmark (SDU), dVRK Surgical Robotics | October 2025 - May 2026

- Built the full learning-based control pipeline for a bimanual suturing task on the physical da Vinci Research Kit (dVRK), end to end: synchronized data collection, LeRobot dataset conversion, diffusion-policy training, ROS 1 deployment, and closed-loop evaluation on the real robot. (project page, result video)
- Diagnosed a vision-encoder failure with representation probes (decoder feature-inversion adapted from LeWorldModel, Yann LeCun et al.) and improved policy behavior by moving from a ResNet to a fine-tuned DINOv2 encoder with state-dropout.
- Bridged the gap between simulated/theoretical models and real-world execution on a surgical robotics platform.
- Supervised by Associate Professor Iñigo Iturrate. (profile, recommendation letter)
- Technologies: dVRK, ROS 1, PyTorch, Diffusion Policy, DINOv2, Python, C++.

Student Researcher (Part-Time) | University of Southern Denmark (SDU), Surgical Teleoperation | June 2026 - October 2026

- Laid the foundation for teleoperating the lab's UR5 cell, which carries a da Vinci surgical instrument, from a GELLO leader arm: leader firmware and calibration, one-to-one RViz/URSim synchronization, and clutched Cartesian control through the lab's existing IK, UR RTDE and instrument (EUA) controllers.
- Added a safety layer (speed and lag limits, runaway rejection, disengagement on stale feedback or a lost pedal), validated in URSim and a hardware dry run (instrument following the leader, UR arm idle), and handed over to the lab after a short engagement.
- Technologies: ROS 2, Python, Docker, UR RTDE, DYNAMIXEL.

TECHNICAL SKILLS

Programming: C/C++, Python, MATLAB | **Robotics:** ROS 2, ROS, SLAM, Localization, Sensor Fusion, Control Systems, Path Planning, Motion Planning | **Computer Vision:** OpenCV, YOLO, Object Detection | **Robot Learning:** PyTorch, Diffusion Policy, Imitation Learning | **Tools:** GitHub, Git, Linux, Docker | **Soft Skills:** Team Leadership, Project Management

EDUCATION

MSc in Engineering, Robot Systems (Advanced Robotics Technology) | University of Southern Denmark, Odense | August 2025 - June 2027 (expected)

- Full scholarship recipient.
- Thesis (with Saha Robotik): Language-Conditioned Mobile Manipulation for Restaurant Table Clearing: Adapting Vision-Language-Action Models to a Bimanual Mobile Service Robot.

Bachelor of Science in Mechatronics Engineering | Kocaeli University, Turkey | August 2019 - October 2024

- GPA: 3.19 / 4.00 (Honor Student).

CERTIFICATIONS

IELTS Overall Band Score: 6.5 (Listening: 7.5, Reading: 7.0, Writing: 5.5, Speaking: 5.5).

VOLUNTEER EXPERIENCE

Team Lead → Mentor | KOU-Mekatronom Autonomous Vehicle Team, Kocaeli University | November 2021 - Present

- Led a 20-member student team developing autonomous vehicle systems for international competitions.
- Selected and led a five-person squad to the Bosch Future Mobility Challenge on-site final, the only Turkish finalist (Top 24 of 160 global teams). (project page, code, video)
- Placed 6th at Teknofest Robotaxi 2023, up from 19th in 2022.
- Managed project planning, resource allocation, and technical development for the autonomous-driving stack.
- Open-sourced project code and tutorials to support future cohorts.
- Led the team through 2024 and continue to mentor since then; as of 2026 still a mentor, with limited time on critical points.
- Technologies: C++, Python, MATLAB, OpenCV, YOLO, ROS, Computer Vision, Path Planning.

Volunteer Teacher | Bambu Voluntary Teaching Platform | January 2021 - August 2021

- Provided STEM tutoring for middle- and high-school students during the COVID-19 pandemic.

REFERENCES

Professor Metin Aydın | Kocaeli University | Email: metin.aydin@kocaeli.edu.tr

Asst. Prof. Suat Karakaya | Kocaeli University | Email: suat.karakaya@kocaeli.edu.tr | Recommendation letter

Jacob Pilegaard | R&D Software Manager, Universal Robots | [Contact information to be provided upon request]

Assoc. Prof. Iñigo Iturrate | SDU (dVRK supervisor) | Email: inju@mmmi.sdu.dk | Profile | Recommendation letter

Oğuzhan Köse | CTO, Saha Robotik | Contact details available upon request

Murat Ayrancı | Founder & CEO, Saha Robotik | Contact details available upon request